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Date
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ASSIGNMENT - 2

Data Communication and Networks End Term Examination (2019)

Q1

Ans (a) A number of design issues exist for the layer to layer approach of data computer networks. Some of the main design issues are as follows:-

- **Reliability:-** Network channels and components may be unreliable, resulting in loss of bits while data transfer. So, make sure that the information transferred is not distorted.
- **Scalability:-** Networks are continuously evolving. The sizes are continually increasing leading to congestion. So for better design networks are scalable and can accommodate such additions and alterations.
- **Addressing:-** At a particular time, innumerable messages are being transferred between large no. of computers. So that each layer can identify the sender & receiver of each message.
- **Flow Control:-** If the data is produced by sender is higher than the rate at the data

is received by receiver, there are chances of overflowing the receiver. So, a proper flow control mechanism to be implemented.

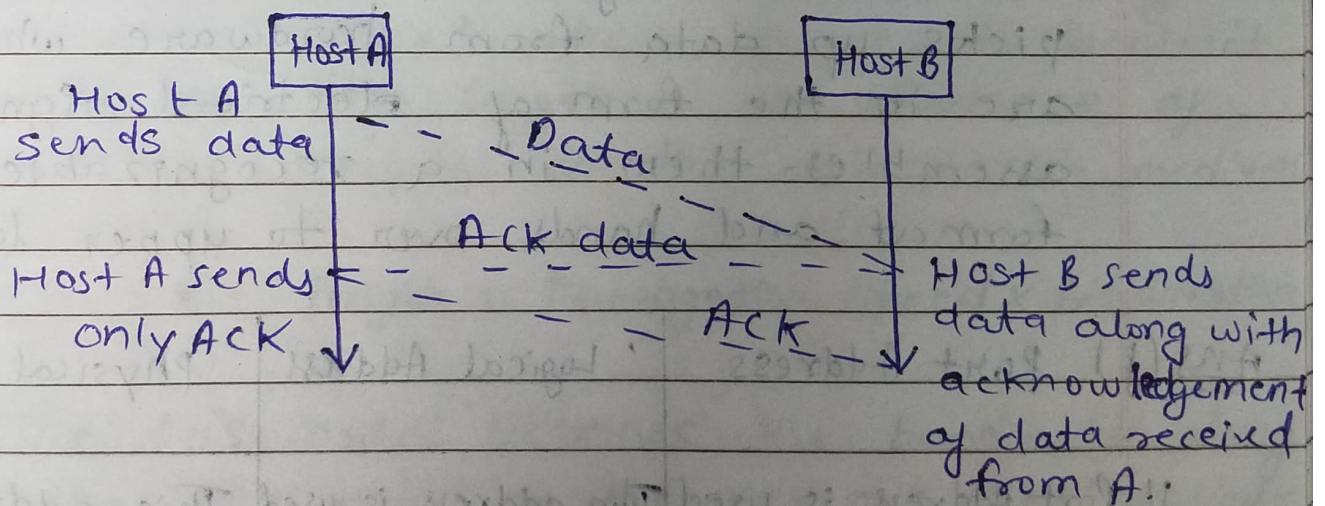
Routing:- There may be multiple paths from the source to destination. Routing involves choosing an optimal path all possible path, in terms of cost & time. There are several routing algorithms that are used in network system.

(b) The four fundamental characteristics of a Data Communication Network are:-

- 1) Delivery:- The system must deliver data to the correct destination. Data must be received by device or user and only.
- 2) Accuracy:- The system must deliver the data accurately. Data that have been altered in transmission & left uncorrected are unusable.
- 3) Timeliness:- The system must deliver data in a timely manner. Data delivered late are useless. In case of video & audio, timely delivery means delivering data as they are produced in the same order that they are produced without delay. This is called real-time transmission.

4) Jitter :- Jitter refers to the variation in the packet arrival time. It is uneven delay in the audio or video packets.

Ans(c) Consider a two way transmission b/w host A and host B. When host A sends a data frame to B, then B does not send the acknowledgement of frame sent immediately. The acknowledgement is delayed until the next data frame of host B is available for transmission. The delayed acknowledgement is then attached to the outgoing data frame of B. This process of delaying acknowledgement so that it can be attached to outgoing frame is called piggybacking.



Ans(d) The most important criteria are performance, reliability and security.

Performance :- Performance of the network depends upon number of users, type of

transmission medium; and the capabilities of the connected h/w and the efficiency of the s/w.

Reliability :- Reliability is measured by frequency of failure, the time it takes a link to recover from the failure and the network's robustness in a catastrophe.

Security :- Security issues include protecting data from unauthorized access & viruses.

Ans (e) Data link layer is responsible for converting data stream to signal bit by bit and send to that over the underlying hardware. At the receiving end, data link layer picks up data from hardware which are in the form of electrical signals, assembles them in a recognizable frame format and hands over to upper layer.

<u>Ans (f)</u> Port Address	Logical Address	Physical Address
1) Port Address is used to identify the particular application running on the destination m/c.	Two address is used by network layer to identify a particular network among the network.	This address is used by data link layer.

2) Each application run with a port no. on the computer	An IP address of the system is called logical Address	The address of NIC is called physical Address.
3) This port no. for application is divided by the kernel of the OS	This address is the combination of Net IP & Host IP	This is specified by the manufacturer company of the card.

Ans (g) Bit interval is the time required to send one signal bit of data. Whereas Baud rate is the number of single unit transmitted per second. It is always less than or equal to bit rate.

Bit rate is the number of bits transmitted per second. It depends on the frequency of the material which is used to transmit this data. Bit length is the distance of one bit occupies on the transmission medium.

$$\text{Bit length} = \text{Propagation speed} \times \text{Bit duration}$$

Ans (h) The QoS model reserves the bandwidth along a specific path on the network. The application ask the network's resource reservation for themselves and parallelly the network devices monitor the flow of packets to make sure network resources can accept packets.

Ans (i) Classless Addressing is an improved IP addressing system. It makes the allocation of IP addresses more efficient. It replaces the older classful addressing system based on classes. It is also known as classless inter domain routing (CIDR).

Ans (j) A checksum is a small sized block of data derived from another block of digital data for the purpose of detecting errors that may have been introduced during its transmission or storage. Checksums are often used to verify data integrity but are not relied upon to verify data authenticity.

Unit - 2

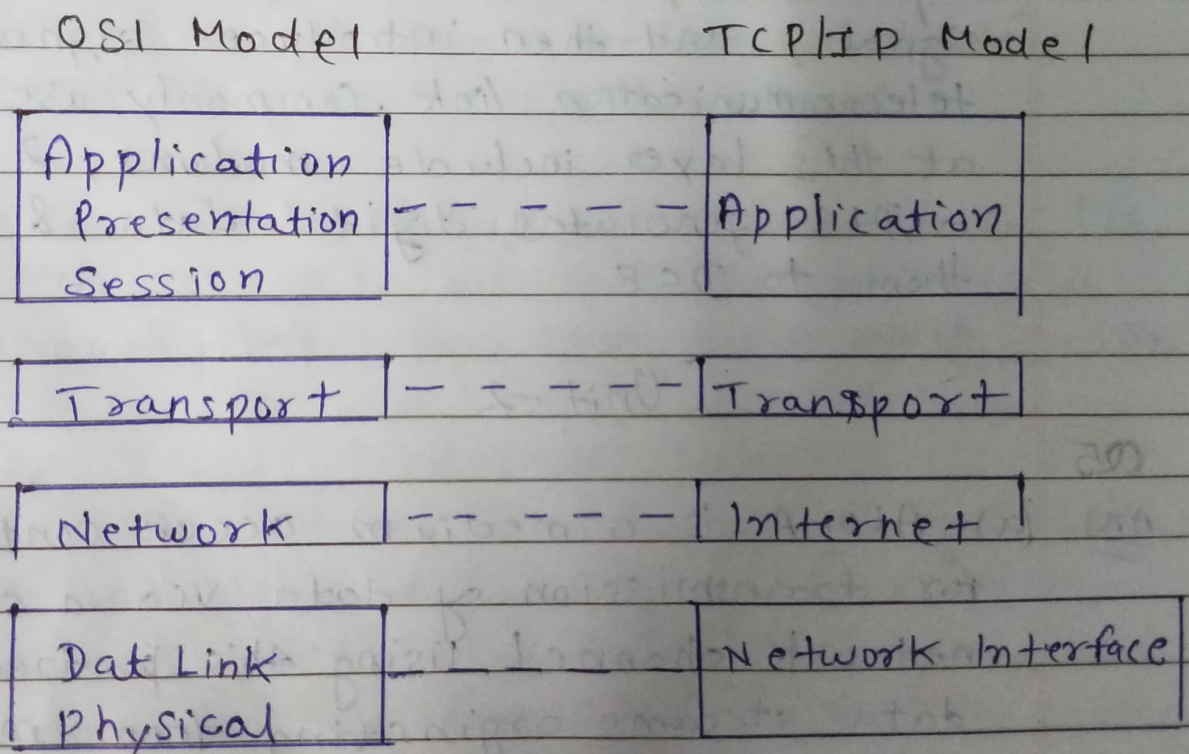
Q2

Ans (a) TCP/IP and OSI are the two most widely used networking models for communication. There are some similarities & dissimilarities between them. One of the major difference is that OSI is a conceptual model which is not practically used for communication whereas, TCP/IP is used for establishing a connection and communicating through the network.

The OSI model mainly emphasis on the services, interfaces and protocols, make a clear distinction between these. Conversely

The TCP model is not able to distinctly describes these concepts. furthermore, the TCP/IP enables only connectionless communication mode in the network layer but both modes in the transport layer.

When it comes to the OSI model, it supports connectionless and connection oriented communication over the network layer but, in the transport layer connection oriented communication is merely allowed.



2/b) Data Terminal Equipment (DTE) :- It includes any unit that function as either as a source of or as a destination for binary digital data. At physical layer it can be a terminal, microcomputer,

computer, printer, fax machine or any other device that generates or consumes digital data. DTEs do not often communicate information but need an intermediary to be able to communicate.

Data Circuit Equipment (DCE) : It includes any functional unit that transmit or receives data in form of an analog or digital signal through a network. At physical layer, a DCE takes place data generated by a DTE, converts them to an appropriate signal, and then introduces signal into telecommunication link. Commonly used DCE's at this layer include modems, & any network a DTE generates digital data & passes them to DCE.

Unit - 2

Q5

Ans (a) ALOHA is a medium access control protocol for transmission of data via a shared network channel. Using this protocol, several data streams originating from multiple nodes are transferred through a multi-point transmission channel. There are two types of ALOHA protocols - Pure ALOHA and Slotted ALOHA.

Slotted ALOHA is different from Pure ALOHA as

- 1) In pure ALOHA, whenever a station has data to send it transmits it without waiting whereas in slotted ALOHA a user wait till the next time slot begins to transmit the data.
- 2) In slotted ALOHA the time is discrete and divided into slots whereas in pure ALOHA the time is continuous.
- 3) The time of sender and receiver in pure ALOHA is not globally synchronized whereas the time of send and receiver in slotted ALOHA is globally synchronized.
- 4) The maximum throughput occurs at $G = 1/2$ which is 18%. whereas the maximum throughput occurs at $G = 1$ which is 37%.

Ans 5(b) High Level Data Link Control (HDLC) :
 Generally uses term "frame" to indicate and represent an entity of data or a protocol of data unit often transmitted or transferred from one station to another station. Each and every frame on link should begin and end with flag sequence field. Each of frames in HDLC includes mainly 5 fields. It begins with a key flag field, an address field, a control field, an information field, a frame check

Sequence field, and an ending flag field.

- **Flag field:-** The flag field is generally responsible for initiation and termination of error checking. In HDLC protocol, there is no start and stop bits. So the flag field is basically using delimiters 0x7e to simply indicate beginning and end of frame.
- **Address field:-** The address field generally includes HDLC address of secondary station. It helps to identify secondary station will sent or receive data frame.
- **Control field:-** HDLC generally uses this field to determine how to control process of communication. The field is a 1-2 byte segment of frame generally requires for flow and error control.
- **Information field:-** The field usually containing data or information of user. Sender is transmitting to receiver in an frame (I) and network layer or management information in U-frame. It also consists of user's data and is fully transparent.
- **Frame Check Sequence (FCS):-** FCS is generally

used for identification of errors i.e. HDLC error detection.

This field can either contain 2 byte or 4 bytes. This field is a total 16 bit that is required for error detection.

Unit -3

Q7

Ans(a)

Distance vector Routing Algorithm.

Distance vector algorithm measures the distance by the number of routers a packet has to pass, one router counts as one hop. It also takes network latency and other factors that influence traffic on a given route. To determine the best route across a network routers, on which a distance vector protocol is implemented exchange information with one another, usually, routing tables plus hop counts for destination networks and possibly other traffic information. Distance vector routing algorithm uses the bellman ford algorithm to calculate the best route. The term distance vectors refers to the fact that the protocol manipulates vectors of distances to other nodes in the network.

Link state Routing Algorithm:- The link state algorithm is performed by every switching node in the network. The basic concept of

link state routing is that every node constructs a map of connectivity to the network in the form of a graph, showing which nodes are connected to which other nodes. Each node then calculates the next best logical path from it to every possible destination in the network. Each collection of best paths will then form each node's routing table.

Ans 7 (b) Classful Addressing (in IPv4)
classful addressing splits the available address space into five classes A, B, C, D, & E. IPv4 addresses are represented using 32 bit addresses. The 32 bit IPv4 address is also referred to as the 4 byte address or 4 octet address. So we can include that the address space of IPv4 is 2^{32} which is equal to 4,294,967,296.

Generally the IPv4 addresses are expressed using the binary notation or dotted decimal notation or hexadecimal notation. The few bits of binary notation of IPv4 addresses recognize the class of the addresses whereas, in dotted decimal notation of IPv4 address the value of the first byte recognize the class of the address. The classful addressing concept divides the address space into a fixed number of blocks.

& each block has a fixed number of hosts. In IPv4 addresses of class A, B & C the first part of the address is considered as the net id and the second part of the address is called host-id. The size of these parts varies with the classes.

Unit-4

Q9

Ans (i) Domain Name System (DNS)

An application layer protocol defines how the application processing running on different systems, pass the messages to each other. It is a directory service that provides a mapping between the name of a host on the network and its numerical address.

Each node in a tree has a domain name, and a full domain name is a sequence of symbols specified by dots. DNS is a service that translates the domain name into IP addresses. This allows the users of networks to utilize user friendly names of when looking for other hosts instead of remembering the IP addresses.

For ex - suppose the FTP site at EduSoft had an IP address of 132.147.165.50, most people would reach this site by direct ftp.eduSoft.com.

(iv) File transfer Protocol (FTP) :-

File transfer Protocol is a set of rules that computer follow for the transferring of files from one system to another over the internet. It may be used by a business to transfer files from one computer system to another, or websites may use FTP to upload or download files from a website's server. It is a way to download, upload and transfer files from one location to another on the internet & between computer systems. Many file transfer protocol clients are available for free to download, although most websites that offer downloads already have the FTP built in.

Once you have accessed the FTP server through FTP client, you can now transfer files. Not all public FTP servers require you to sign in because some servers enable you to access them anonymously.

(v) World wide web (WWW) :-

It is also known as web, is a collection of websites or web pages stored in web servers and connected to local computers through Internet. These websites contain text pages, digital images, audios, videos, etc. User can access the content of these sites from any part of the world over

the internet using their devices such as computers, laptops, cell phones etc. The WWW, along with Internet, enables the retrieval and display of text media to your device.

The building blocks of the web are web pages which are formatted in HTML and connected by links called "hypertext" or hyperlinks and accessed by HTTP. These links are electronic connections that related pieces of information so that users can access the desired information quickly.

A web page is given an online address called a Uniform Resource Locator (URL). A particular collection of web pages that belong to a specific URL is called a website, e.g, www.facebook.com, www.google.com etc.